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Paleomagnetism of Late Hadrynian Fourchu Gabbro, Southern Cape Breton Island, Nova Scotia

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A paleomagnetic study was carried out on three gabbroic intrusives in the Avalonian terrane of southeastern Cape Breton Island. Thirty five samples (96 specimens) from seven sites were collected for this purpose. Thermal and/or alternating field treatments were performed. Four components of magnetization were detected. One component coincides with the present Earth's field direction. A second component (A) oriented south-southeast with shallow to intermediate inclination (165, +41) is interpreted as representing a Siluro-Devonian overprint. A third component (B) yields a mean direction to the southeast and an intermediate inclination (147, −54). This component is not observed in North American rock units of Devonian age or younger; it is thus bracketed in the Early Devonian–Late Hadrynian interval. Based on magnetic properties, mineralogical and geological grounds, an Ordovician-Cambrian age of acquisition is attributed to this component of magnetization. A fourth component characterized by a low unblocking temperature and median destructive field has a mean direction to the northeast and an intermediate inclination (54, +47). This uncommon direction is inferred to represent an incompletely isolated magnetization. Compaged to low southern paleolatitudes for the Late Hadrynian period of cratonic North America, the paleolatitudes of the Avalonian terrane are slightly higher but this is not really significant. A sea at the most 1000 km wide may have separated North America and Avalon at that time. A comparison of the paleolatitude of Avalon and those of Late Hadrynian Atlantic Ocean borderlands indicates that Avalon could have belonged to North Africa at that time. The Avalonian apparent polar wander path in the Devonian–Late Hadrynian interval still remains uncharted. The collision of Avalon with North America occurred most probably at Middle Devonian time.

1. Introduction

The Avalon Zone

Cape Breton and northern Nova Scotia are part of the Avalon Zone. Many geoscientists have attempted to define the extent, origin and type of plate motion for displaced terranes in the northern Appalachians (WILLIAMS, 1978, 1979; LEFORT and HAWORTH, 1978; HAWORTH and LEFORT, 1979; BRÜCKNER, 1980; PIQUÉ, 1981; WILLIAMS and HATCHER, 1983). Geological arguments for a possible displacement of Avalon are based on 1) dissimilar pre-Late Paleozoic geologic records of the Avalon Zone and other zones of the northern Appalachians, 2) different Early Paleozoic faunas for Avalon and cratonic North America (WILSON, 1966; MCKERROW and ZIEGLER, 1972; WHITTINGTON and HUGHES, 1972; MCKERROW and COCKS, 1976; COCKS and

FORTEY, 1982; NEWMAN, 1984), 3) actual fragments of probable Paleozoic crust between Avalon Zone and cratonic North America (ZEN and PALMER, 1981; WILLIAMS and HATCHER, 1983; RAST and SKEHAN, 1983; WILLIAMS, 1984). Precambrian rocks similar to the ones in the Avalon Zone occur throughout the Appalachian orogen in Newfoundland, Nova Scotia, New Brunswick, eastern Massachusetts, southeastern United States (Carolina State Belt) (SCHENK, 1972, 1978; RANKIN, 1976; WILLIAMS, 1979) and on the eastern side of the Atlantic Ocean (Wales, Brittany, Iberian Peninsula, Czechoslovakia, Turkey and northeast Africa).

Various tectonic histories have been proposed for the origin of the Avalon mini-continent. The tectonic models include volcanic island arc (HUGHES and BRÜCKNER, 1972), ensialic island arc (RAST *et al.*, 1976), subduction and Basin and Range type rifting (PAPEZIK, 1972; SCHENK, 1978), continental extension and rifting (RANKIN, 1976), transitional tectonic regime (KING, 1982) resulting from continental margin orogenesis (RAST *et al.*, 1976) and intraplate orogenesis (STRONG *et al.*, 1978), Pan-African terrane predating the opening of an Early Paleozoic proto-Atlantic ocean (O'BRIEN *et al.*, 1983) distinct geological province or "suspect terrane" (WILLIAMS and HATCHER, 1982) moving independently from other Atlantic borderlands.

As it is not yet possible to relocate the Avalon Zone unambiguously (WILLIAMS and MAX, 1980), a first problem to be solved is related to the Early Paleozoic position of the Avalonian terranes, e.g. is Avalon part of Cratonic North America, of another large tectonic plate, an independent block, etc.? A second problem concerning the history of Avalon is related to the timing of collision (juxtaposition) with cratonic North America. Problems related to plate origin and dynamics cannot generally be resolved with geological information alone; this is where geophysical tools become useful. Amongst the variety of geophysical methods, paleomagnetism is particularly well suited to determine the provenance of a (micro) plate and to quantify its relative displacement as a function of time. In principle, drift motions are extractable from the paleomagnetic record but in practice it is a difficult task; the timing of acquisition of remanences compared across hypothetical collision sutures or fault boundaries are generally hard to establish. On geological grounds, the inferred timing of motions along the fault separating the Avalon and Gander zones in Nova Scotia range from Late Precambrian (KENNEDY *et al.*, 1982) to Early Carboniferous for the coalescence of the two zones (ELIAS and STRONG, 1982). The Minas geofracture (a dextral transcurrent fault) separating Meguma and Avalon zones is dated at mid-Devonian time (KEPPIE, 1982).

2. Previous Paleomagneticstudies on the Avalon Zone

Paleomagnetic results from Avalon in Canada were first reported in Upper Hadrynian rock units in Newfoundland (NAIRN *et al.*, 1959) and in Lower Cambrian and Upper Devonian lavas and sediments in New Brunswick (BLACK, 1964). Newer Avalonian results were obtained from Lower Ordovician red sandstones in Newfoundland (RAO, 1970; BUCHAN and HODYCH, 1982; ÖZDEMIR and DEUTSCH, 1984), Siluro-Devonian gabbros (ROY *et al.*, 1979) and sedimentary rocks (ROY and ANDERSON, 1981) from New Brunswick, from Carboniferous redbeds in New Brunswick (ROY and PARK, 1974), in Newfoundland (KENT, 1982) and in Nova Scotia (SCOTESE *et al.*, 1984). Other Avalonian paleomagnetic data include the Cambrian to Devonian granitic intrusives from Cape Breton (RAO *et al.*, 1981), the Siluro-Devonian mafic dykes (SEGUIN *et al.*,

1982a) from Newfoundland, the Early Silurian Dunn Point volcanics (VAN DER VOO and JOHNSON, 1985) from Nova Scotia and Upper Hadrynian as well as Late Devonian rock formations from the Burin Peninsula (IRVING and STRONG, 1985) in Newfoundland. A summary of all these paleomagnetic poles is presented in Table 2 of the study of Early Cambrian redbeds (RAO *et al.*, 1986) from Cape Breton. Recent paleomagnetic results include the following studies: 1) Upper Hadrynian Finn Hill sequence (SEGUIN, 1985), 2) prophyrites intrusives (SEGUIN and LORTIE, 1986), and 3) Harbour Main Group metabasalts (SEGUIN and GIASSON, 1987) in Newfoundland, 4) Late Precambrian Fourchu Group (JOHNSON and VAN DER VOO, 1986) from Cape Breton, 5) Lower Cambrian Brigus shale (SEGUIN and RAO, 1989) from Newfoundland and 6) Middle Cambrian Bourinot Group (JOHNSON and VAN DER VOO, 1985) from Cape Breton.

The Fourchu gabbro has been studied for paleomagnetic purpose by JOHNSON and VAN DER VOO (1986). In the present study, not only the dikes but also small gabbro stocks (3) were investigated. In addition, the sites of the sampled gabbro dikes and sills are different from those of JOHNSON and VAN DER VOO (1986). The Avalon poles are widely dispersed and no readily evident apparent polar wandering path (APWP) has yet been established. A post-Devonian sinistral displacement of "Acadia" relative to cratonic North America proposed by KENT and OPDYKE (1978, 1979) and VAN DER VOO *et al.* (1979) is unsupported by recent paleomagnetic data (SEGUIN *et al.*, 1985; SEGUIN and FYFFE, 1985; IRVING and STRONG, 1985; KENT and OPDYKE, 1985). KENT (1982) proposed that the Avalon platform shared this hypothetical sinistral displacement; however, IRVING and STRONG (1984a, b, 1985) have shown that the magnetizations used to document this displacement hypothesis were in fact Late Carboniferous-Permian overprints.

In summary, additional paleomagnetic data from Late Precambrian (Hadrynian) and Early Paleozoic are needed to chart the motions of the Avalon Zone. An obvious gap in the record exists in the Hadrynian. We report here the results from the Late Hadrynian Fourchu gabbro in western Cape Breton Island.

3. Geologic Setting

The Proterozoic Fourchu Group of southeastern Cape Breton Island (Fig. 1) consists of sodic felsic pyroclastics, ignimbrites, minor rhyolitic flows, interbedded mafic tuffs, agglomerates and volcaniclastic sediments (WEEKS, 1954; MURPHY, 1977). Agglomerates and felsic tuffs are predominant rock units. Bedding and cross-stratification indicate a subaerial to shallow water environment of deposition. The Fourchu Group is conformably overlain by Eocambrian conglomerates and a fossiliferous Lower Cambrian sedimentary sequence (SMITH, 1978; RAO *et al.*, 1986). It is underlain unconformably by a crystalline complex of probable Archean age (WEEKS, 1954; MURPHY, 1977). The Fourchu Group was correlated with similar volcanosedimentary units in eastern Newfoundland and southern New Brunswick (WEEKS, 1954; RAST *et al.*, 1976; O'BRIEN *et al.*, 1983) which gave radiometric ages (Rb/Sr) in the −630 to −600 Ma range. Deformation is characterized by close folds and an associated penetrative axial planar deformation (MURPHY, 1977). A second foliation is observed locally; it could represent a late Acadian age deformation but WEEKS (1954) concluded that the Paleozoic strata overlying the Fourchu Group had not suffered internal deformation but tilting only. SMITH (1978) inferred a Late Hadrynian age for this deformation which can be

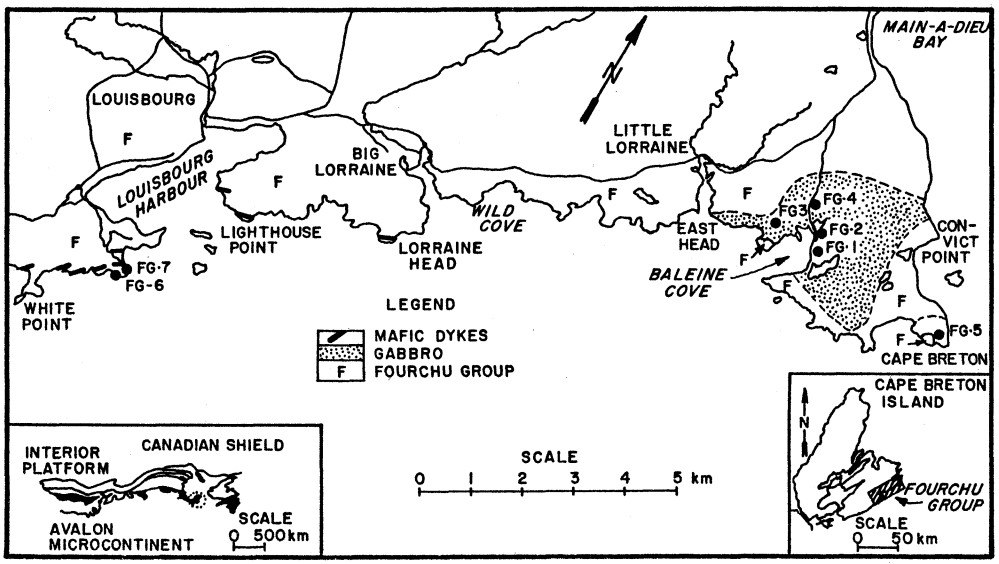


Fig. 1. Geologic setting of Late Precambrian Avalonian rocks in southern Cape Breton. The Fourchu gabbro intrudes the Fourchu Group. The data show the paleomagnetic sampling sites (geology after WEEKS (1954) and MURPHY (1977)).

correlated with similar deformation in other parts of the Avalon Zone (STRONG, 1979). Erosion apparently followed immediately the period of deformation (RAST *et al.*, 1976).

The rocks that intruded the Fourchu Group were divided into four members (MURPHY, 1977): 1) early felsic sheets, 2) mafic sheets, 3) gabbro, and 4) late felsic sheets. Gabbro bodies are found in several localities, e.g. Blackrock Point, Baleine Cove and Cape Breton which constitute the sampling area. The gabbros are dark green and consist of phenocrysts of pyroxene in a plagioclase-rich matrix; their degree of alteration is variable but it is generally higher in the vicinity of the chilled margins. The margins of the smaller gabbro bodies are foliated while the core is massive. In the layered bodies (e.g. Baleine Cove), the Fourchu Group country rocks have been distorted in the immediate vicinity of the intrusion. In general, the contacts are sub-vertical and sharp. Except for very minor occurrences of late felsic sheets, the gabbro intrusives represent the last major igneous event of the area. According to MURPHY (1977), foliation of intrusive margins, evidence of cataclastic deformation and other related observations indicate that the gabbroic bodies were intruded before the Late Hadrynian tectonic event.

4. Paleomagnetic Sampling and Magnetization Measurements

Suitable gabbro exposures could be obtained in the Louisbourg and Baleine Cove areas where we collected 35 oriented hand samples at seven sites (Fig. 1). The direction and moment of natural remanent magnetization (NRM) were measured with a Schonstedt spinner magnetometer (DSM-1). Stepwise demagnetizations of NRM (ROY, 1986) were carried out using (1) an alternating field (AF) demagnetizer built at University Laval (SEGUIN, 1975) and a Schonstedt apparatus (Model GSD-1) in residual fields of less than

10 nT, and (2) a Schonstedt thermal demagnetizer (TSD-1) up to 640° C in residual fields of less than 15 nT. In addition, we employed occasionally a two-stage cleaning technique consisting of AF demagnetization followed by thermal treatment (PARK, 1975; ROY and LAPOINTE, 1976). Between two and five specimens were used to obtain a sample-mean NRM direction. Room-temperature volume magnetic susceptibilities, K, were measured in the Earth's field (H=0.05 mT) with a Cenco MS-3B susceptibility bridge unit.

Individual specimen NRM directions of Fourchu gabbro, although characterized by a wide scatter, have a pronounced deviation away from the present Earth's field (PEF) D=336°, I=+71°) at the site localities. This indicates that the gabbro carries some ancient remanence, especially since some directions of remanence of the specimens (15%) located close to PEF were found to move away from these initial directions during subsequent AF and/or thermal treatments. The NRM directions are characterized by a NW-SE trend (Fig. 2(a)); the intensities of NRM (Jn) vary from 4×10^{-4} to 3×10^{-1} Am^{-1} with the value of Koenigsberger ratio $Qn=(Jn/KH)$ between 1 and 30 but with predominant ratios between 1 and 8 (Fig. 2(b)). The smallest values of Jn are found in sites FG.1, FG.2 and FG.3 and the largest in sites FG.6 and FG.7.

5. Paleomagnetic Results

5.1 AF demagnetization

At least one specimen from each of 35 samples was subjected to AF demagnetization in steps of 5 mT in the 5–30 mT range and in steps of 10 mT in the 30–100 mT range. The AF demagnetization results were subsequently analysed using orthogonal vector diagrams (ZIJDERVELD, 1967) and stereograms in parallel with decay curves of normalized NRM intensity. Multivectorial component magnetizations were most commonly separated by using orthogonal vector diagrams. A shift in the mean direction away from the PEF at all sites indicates complete or partial removal of minor viscous remanence. For both within and between site localities, the agreement in NRM directions is much improved after the removal of the viscous component in a field of 5–15 mT (Figs. 3(a), (b) and (c)). In addition to the PEF, two components (A, B) of magnetization were isolated. Component A is the most common; it is found in 32 out of 35 samples and both normal (specimen FG 2-4.2, Figs. 3(a) and (b)) and reverse (specimen FG 4-1.2, Figs. 3(a) and (b)) polarities are observed. It is characterized by a southern declination and an intermediate positive inclination (Fig. 4(a)). Its median destructive field ranges from 5 to 40 mT (mean: 15 mT) and at about 30–35 mT, the intensity of magnetization decreases to 10% of its original value. When the PEF is present, it always overprints component A (FG 2-4.2, Figs. 3(a) and (b)). Figure 3(c) shows an example of orthogonal vector diagram indicating the overprinting relation between the PEF and component A (sample FG 2-1). Orthogonal vector diagram and vector subtraction are the best methods to extract the individual components. The PEF overprinted on component A is readily observed in the grouping of magnetization directions on a stereonet projection during stepwise AF demagnetization or through the linearity of orthogonal vector diagrams. Component B is the second most frequent component; it is observed in 24 out of 35 samples (FG 3-3.2, Fig. 3(a)). Components A and B are found together in 35 specimens. Component B (Fig. 4(b)) has a magnetic coercivity higher than that of component A as shown in Fig. 3. It is characterized by a median destructive field of 40 to 100 mT or more (mean: 55 mT) except when it is overprinted by components A or C. A single polarity (normal) is found

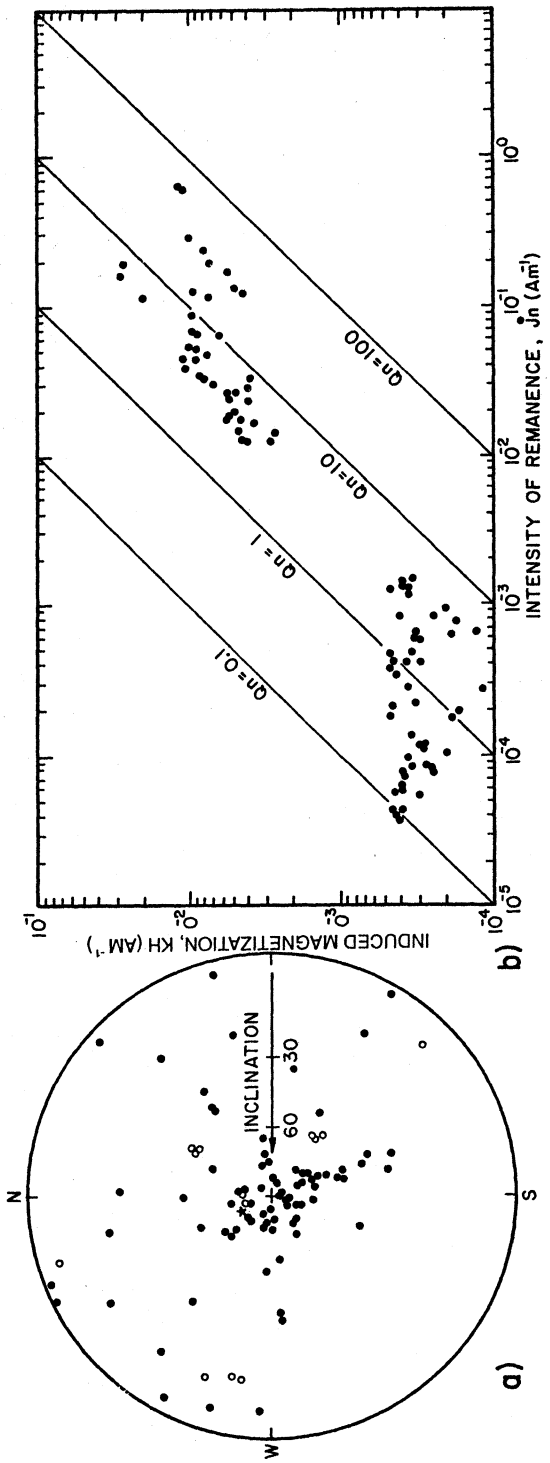


Fig. 2. Individual sample NRM directions on Wulff net for sites FG.1 (left). The solid circles represent projections on lower hemisphere. The star represents the present Earth's field (PEF) at the sampling locality. Plots of intensity of NRM (Jn in Am^{-1}) units against the induced magnetization (KH in SI units) with resulting slope lines indicating the values of Koenigsberger ratio Qn (right) for all samples.

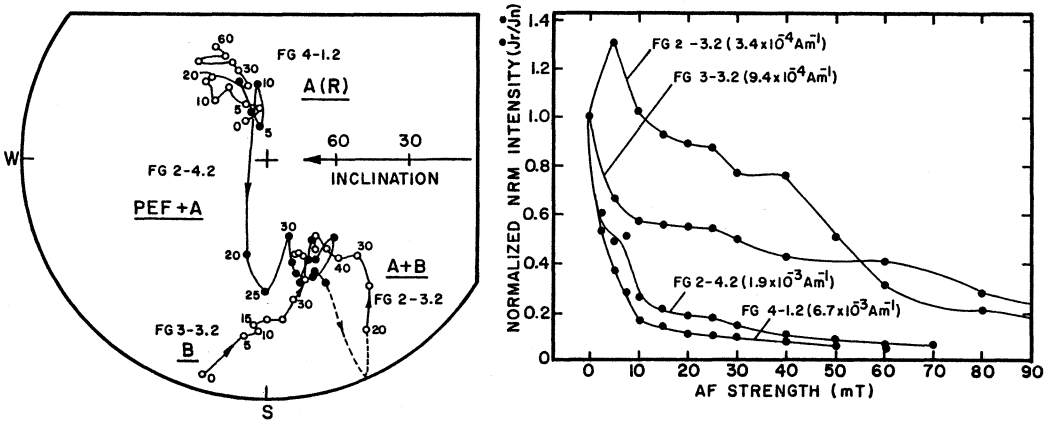


Fig. 3(a). Wulff projection of directions of magnetization (left) of four specimens from sites FG.2, 3 and 4 after stepwise AF demagnetization. The number next to solid (open) circles represent the demagnetization steps in mT as indicated. The open circles represent projections on upper hemisphere. The respective AF decay curves are shown on the right side. Specimen FG.2-4.2 is a typical example of PEF overprinting component A, specimen FG.2-3.2: component A overprinting component B, specimen FG.4-1.2: component A (reverse polarity), specimen FG.3-3.2: component B.

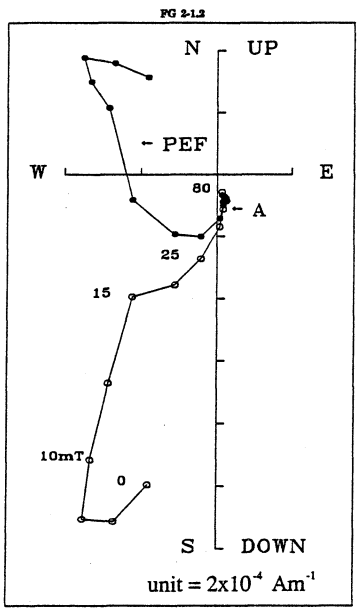


Fig. 3(b). ZIJDERVELD (1967) diagrams showing the projection of magnetization vectors onto the horizontal plane (closed circles) and the vertical plane (open circles). The present Earth's field (PEF) is superimposed on component A. Sample FG.2-1.2 shows the extraction of component A (SSE, intermediate to steep positive) by AF demagnetization.

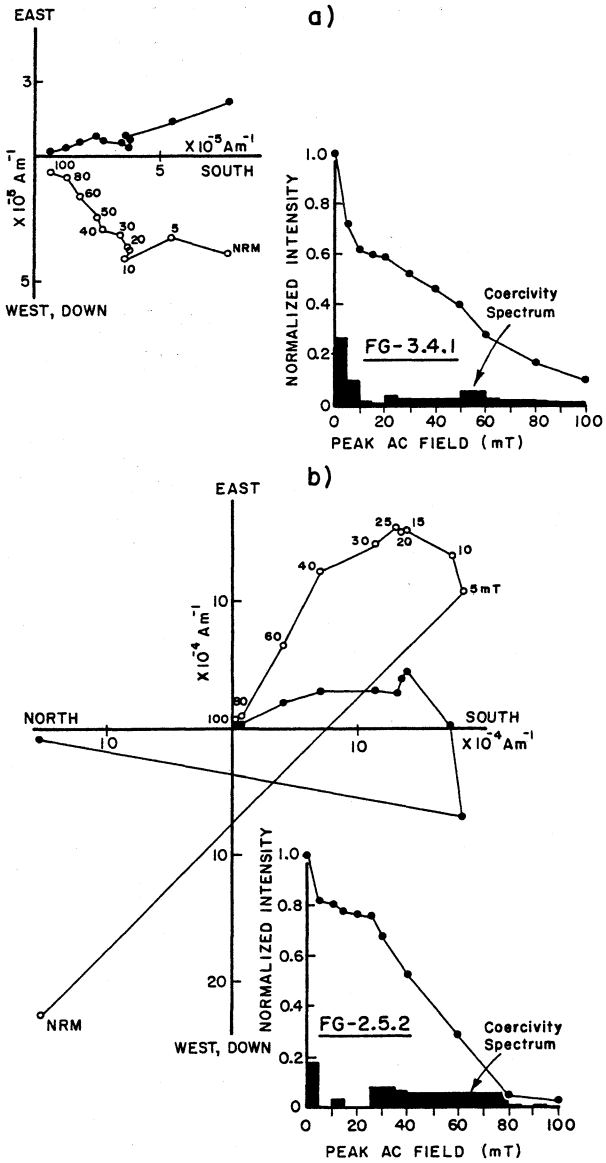


Fig. 4. ZIJDERVELD (1967) diagrams showing the projection of magnetization vectors onto the horizontal plane (closed circles) and the vertical plane (open circles) (a) Sample FG-3.4.1 shows a typical example of extraction of the A component (SE, intermediate positive), (b) Sample FG-2.5.2 shows a typical example of the extraction of the B component (SE, intermediate to steep negative) by AF treatment. The median destructive field and coercivity spectrum are shown in (a) and (b).

in this component which is characterized by a 150° declination and an intermediate negative inclination (Fig. 7).

5.2 Thermal demagnetization

Fifty two (52) specimens were thermally demagnetized at the following steps: 150, 300, 400, 450, 500, 525, 550, 570, 590, 610, 630 and occasionally 660° C. The dispersion of the NRM directions is substantially reduced after the removal of a viscous component in the temperature range: 20–450° C (specimen FG 5-5.1, Fig. 5(a)). Three additional components of magnetization (A, B and C) were isolated. Component A (mixed polarities) is the most frequent (Fig. 6(a)). This southern direction with intermediate positive inclination is characterized by an unblocking temperature (T_{UB}) ranging from 450 to 570° C (mean: 535° C). This component is generally overprinted by the PEF and occasionally component C. Component B (SSE, intermediate negative inclination) is the second most frequent component (Fig. 6(b)); its T_{UB} ranges generally from 570 to 630° C and it is characterized by a single polarity. Component B may be observed as a single component or as a component overprinted by component A (FG 5-4.3, FG 1-3.1, Figs. 5(a) and (b)) or of component C (FG 5-2.1, FG 5-3.2, Figs. 5(a) and (c)). Component C is defined by six points between 150 and 500° C on Fig. 5(c).

Component C is rather uncommon; it is observed in only 9 samples from 4 sites and extracted exclusively by thermal demagnetization (FG 5-2.1, Figs. 5(a) and (c)). Except for 2 specimens, it is associated with component B (FG 5-2.1, Figs. 5(a) and (b)) and is overprinted by a single polarity and its T_{UB} ranges from 300 to 500° C. The site-mean directions of components A and B isolated by AF demagnetization and thermal treatment are similar. Low median destructive fields and unblocking temperatures lower than 570° C in component A indicate that the magnetic carrier is (titano)magnetite. Intermediate to high median destructive fields in component B suggest that both (titano)magnetite and (titano)hematite are the magnetic carriers. High T_{UB}s suggest that the magnetic carrier is (titano)hematite. The presence of (titano)magnetite and (titano)-hematite was confirmed by reflected light microscopic observations (exsolution lamellae). The small amount of these lamellae indicates that the titanium content of hematite is quite low. This could be explained by a slow cooling which would allow the hematite formed from high temperature oxidation of (titano)magnetite to exsolve into titanium-rich (ilmenite) and titanium-poor (hematite) phases. The origin and age of hematite can possibly be inferred from the textures and assemblages. Hematite intergrown with ilmenite in exsolution lamellae probably forms at high temperature during initial cooling (high temperature oxidation). However, other forms of hematite replacing primary (titano)magnetite in ferromagnesian silicates, or fine-grained hematite along grain boundaries could form anytime after the intrusion of the gabbro. Most of the (titano)hematite is associated with the (titano)magnetite but some of it is also found along cleavages and fractures of the pyroxenes and amphiboles. The low T_{UB}'s is of component C suggest that (titano)magnetite is the carrier of this component.

6. Statistical Analysis of the Paleomagnetic Results

The individual remanence directions were obtained through the analyses of ZIJDERVELD (1967) diagrams and vector substraction. The mean-site directions of component A for all 7 sites is 165, +41 (N) ($\alpha_{95}=4.6°$, $k=176.1$), that of component B (7

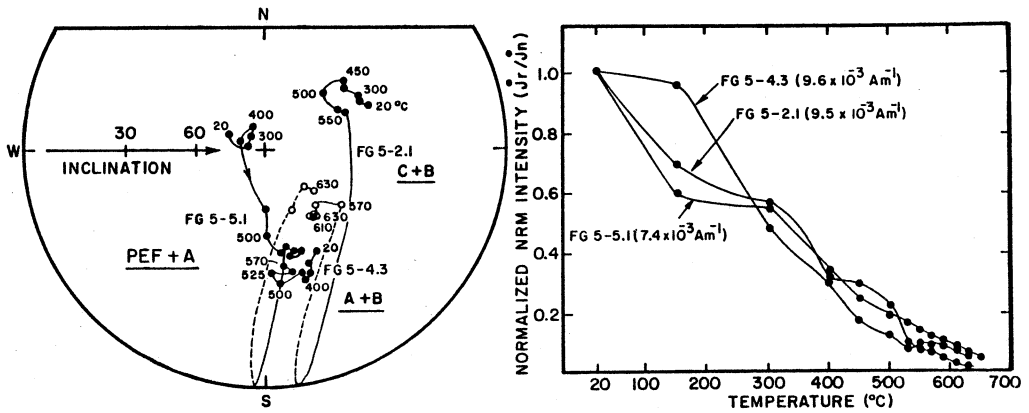


Fig. 5(a). Same projection as for Figs. 2 and 3. Directions of magnetization (left) of three specimens of gabbro after stepwise thermal demagnetization. The respective decay curves are shown on the right side. Specimen FG.5-5.1 is a typical example of the superposition of the PEF on component A, specimen FG.5-4.3, an example of overprinting of component A on component B, and specimen FG.5-2.1, a typical overprinting of component C on component B. The number next to the circles indicate the demagnetization steps in °C. The respective thermal decay curves are shown on the right side.

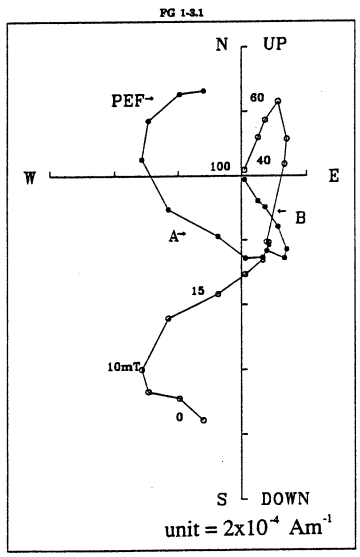


Fig. 5(b). Orthogonal vector diagrams in the horizontal (closed circles) and vertical plane (open circles). Sample FG.1-3.1 shows the extraction of component B (SE, intermediate negative inclination) after removal of component A and the PEF in the course of AF demagnetization.

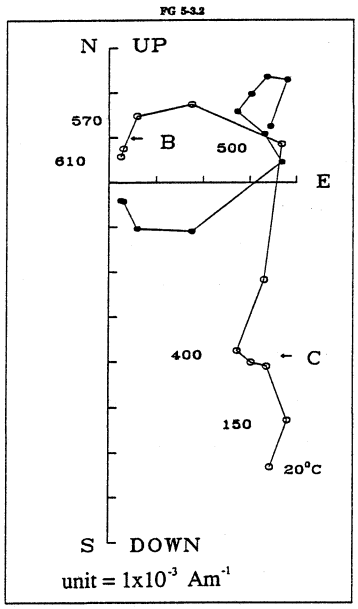


Fig. 5(c). Same as for Fig. 5(b). Sample FG.5-3.2 shows the extraction of component B (SE, intermediate to steep negative inclination) after removal of component C (NE, steep positive) in the course of thermal treatment.

sites as well) is 147, −54 (N) (α_{95}=6.4, k=91.2) and finally that of component C (4 sites only) is 54, 47 (N) (α_{95}=8.6, k=116.3), this last component is less frequently observed. The sample mean directions for each component are shown in Table 1. Figure 7 shows the individual sample directions of magnetization for components A, B and C. The corresponding paleopoles for components A, B and C are 135°E, 19°N (dp=3.4°, dm=5.6°), 196°E, 63°N (dp=6.3°, dm=8.9°), 34°E, 44°N (dp=7.2°, dm=10.9°).

7. Interpretation of the Results

The most difficult task for interpreting the paleomagnetic results concerns the establishment of a sequence of acquisition of magnetization. Criteria of varying importance are used for this purpose, these include: frequency (number of samples per component), polarity, T_{UB}, median destructive field, K vs temperature curves, demagnetization methods, radiometric (or paleontological) age, field tests (baked contact, fold and conglomerate), degree of alteration and metamorphism, paleopole and paleolatitude comparison with those of rock units of the same age and from similar terranes, etc. Obviously only a fraction of these criteria can be applied to a specific paleomagnetic study. The type of alteration in the gabbros can be specified to a certain extent. It does not appear to be deuteric but could result from the burial (regional) metamorphism or the oxidation of existing (original) iron oxides. A combination of the above mentioned criteria can hopefully lead to evidences for the interpretation of the nature and age of the components of magnetization.

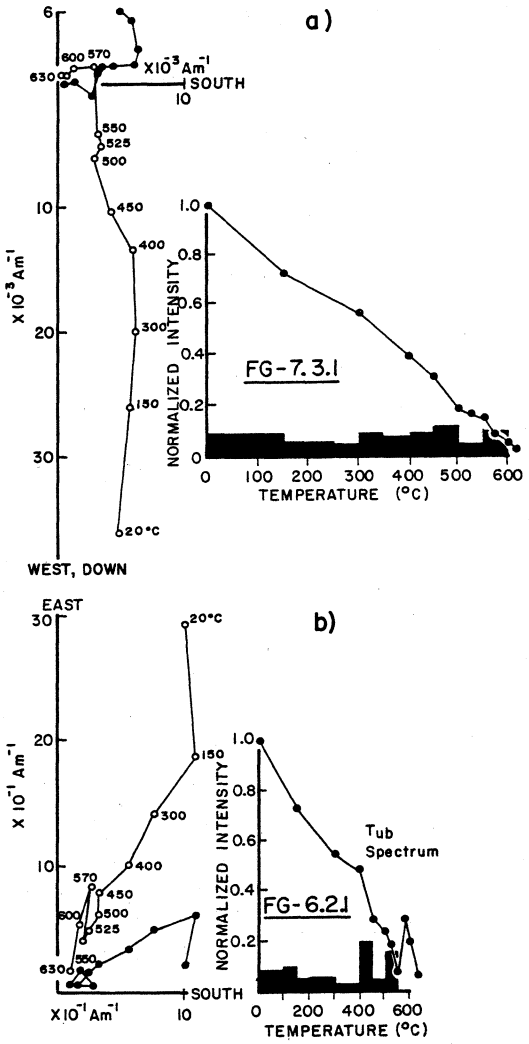


Fig. 6. ZIJDERVELD (1967) diagram. Symbols are the same as in Fig. 4. (a) Sample FG-7.3.1 shows a typical extraction of components A and B with A (SE, intermediate to steep positive and B (SSE, intermediate negative inclination), (b) Sample FG-6.2.1 shows a typical extraction of component B (SE, intermediate to steep negative) by thermal demagnetization. The unblocking temperature spectra are shown in (a) and (b).

In the present study, a first component was clearly identified with the PEF. Second, the low T_{UB}'s, small frequency, impossibility of isolation by AF treatment and higher degree of alteration of the samples in which component C is observed suggest that this magnetization (54, +47) is probably a pseudo-component insufficiently isolated. However, it is noted that a similar component (B; 67, +45) was isolated in the Late Precambrian porphyrites and rhyolites from the Avalon of Newfoundland (SEGUIN and LORTIE, 1986).

Table 1. Site-mean directions of magnetization of Fourchu gabbro.

Site No.	Component A					Component B					Component C				
	N	D (°)	I (°)	α_{95} (°)	k	N	D (°)	I (°)	α_{95} (°)	k	N	D (°)	I (°)	α_{95} (°)	k
FG-1	5	161	38	18.0	18.7	5	148	−53	12.5	38.3					
FG-2	5	166	37	8.4	83.4	5	141	−51	13.1	13.1					
FG-3	5	170	35	17.0	21.1	3	152	−52	32.4	14.1	2	50	48	—	—
FG-4	5	163	46	11.8	61.2	2	167	−57	—	—	2	51	41	—	—
FG-5	5	159	41	15.9	24.1	4	134	−62	21.2	19.3	2	62	44	—	—
FG-6	3	168	46	29.9	15.6	2	154	−42	—	—	3	54	56	8.9	193
FG-7	5	166	49	11.6	44.3	4	146	−53	23.4						
	33	165	41	4.6	176.1	25	147	−54	6.4	91.2	9	54	47	8.9	116

N=number of samples, D=declination, I=inclination, α_{95}=cone of confidence at 95% level, k=precision parameter.

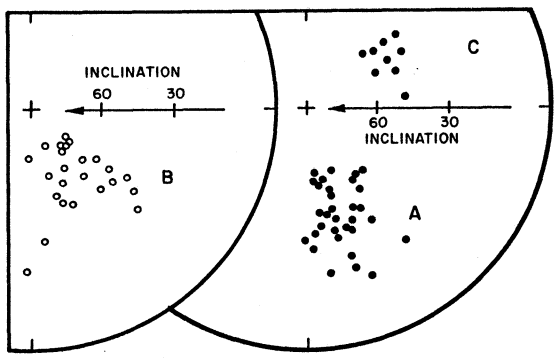


Fig. 7. Wulff projections of sample-mean directions of magnetization obtained for components A, B and C as indicated after both AF demagnetization and thermal treatment. The site-mean directions appear in Table 1.

Two components (A and B) are left and as both cannot be primary, the following scenarios are possible: 1) Both components are secondary, 2) A is primary and B secondary, or 3) B is primary and A secondary. Component A is more frequent, has a smaller median destructive field (MDF) and a lower T_{UB} than component B. Sites FG.1, 2, 3 characterized by a low susceptibility and a low degree of alteration originate from the Baleine Cove area while sites FG.6, 7 having a high susceptibility and higher degree of alteration were collected in the Louisbourg area. Component B is more frequent in sites FG.1, 2, 3 than sites FG.6, 7. Site FG.7 is the only gabbro site close by the Fourchu volcanics and tuffs; it contains inclusions of the country rocks. Even if component B is characterized by a low degree of alteration, a primary age of magnetization remains unproven. Indeed, apparently unaltered grains can acquire a secondary (Viscous) partial Thermal Remanent Magnetization (VpTRM) at relatively low burial temperatures (McELHINNY, 1973; PULLAIAH *et al.*, 1975). So, the relative absence of alteration of the magnetic oxides in component B does not necessarily imply a primary magnetization.

A paleomagnetic study (unpublished) of the volcanosedimentary Fourchu Group has shown that the main cleaned component of magnetization is approximately 170, +65, $\alpha_{95}=12°$ and that the fold test is clearly negative. Consequently, component B found in the Fourchu gabbro is unrelated to this component. A negative fold test of the country rock and a probable positive baked contact test are insufficient arguments to determine the relative age of the two components. However, the post-folding magnetization of the Fourchu volcanics is most likely related to the Avalonian (Cadomian) or Acadian orogenies which affected this region in Late Precambrian and Mid-Devonian times respectively. Indeed, as the inclination of the magnetization is shallow to intermediate at Late Devonian time, the steep inclination (+65) in the Fourchu volcanics suggests that the overprint can date back to very Early Paleozoic time and consequently component B could be as old as Late Precambrian, but this does not necessarily imply that the B component in the Fourchu gabbros is a primary magnetization. In addition, one can question the direction and/or age of magnetization reported for the Fourchu sedimentary and volcanic units. The preliminary direction obtained for these stratified units is reliable even though of limited accuracy; indeed, they were AF and thermally demagnetized and the stability of the characteristic direction of magnetization was well established. In a recently published paleomagnetic study of the Fourchu Group, JOHNSON and VAN DER VOO (1986) obtained a direction of magnetization (their component A) having a comparable inclination ($I=+63°$) but a different declination ($D=312°$). These authors bracket the age of this direction of magnetization between the earliest folding of the rocks (Avalonian event) and the latest Silurian. There are similarities and differences between these two paleomagnetic studies. First the paleolatitudes are the same but the paleopoles are different. Second, the interpretation of the age of this component may be different; both studies agree to a secondary origin of magnetization but the interpretations of the age bracket differ. In this study, the age of this overprint ranges from pre-Devonian to Cambrian while Johnson and Van der Voo bracket this age to different time interval as indicated above.

JOHNSON and VAN DER VOO (1986) isolated two distinct secondary components of magnetization; one, carried by magnetite, has a mean direction of $D=164°$, $I=+16°$ and the second, carried by hematite, has dual polarities and a mean of $D=132°$, $I=+63°$. The characteristic direction of the Fourchu strata reported in this study lies approximately along the great circle between the directions for the two components of magnetization isolated by JOHNSON and VAN DER VOO (1986). These authors observed directions similar to ours but rejected them as likely composite direction resulting from simultaneous removal of two (or more) real components of magnetization. Johnson and Van der Voo note that the vector trajectories during demagnetization of such samples do not decay linearly towards the origin, but are slightly curved, indicating incomplete separation of the components of magnetization. This could suggest that the characteristic direction of the Fourchu strata could be an artifact resulting from incomplete isolation of the components. However, this argument does not hold because in many instances (of this study), the magnetization of component A remains perfectly stable (straight lines on the Zijderveld plots) upon demagnetization like in Fig. 6(a) if the high temperature component (B) is absent.

Even though we are quite certain that the direction of the Fourchu strata reported is correct, the assertion that the direction in the Fourchu strata is too steep to be an overprint of Acadian (Mid-Devonian) age, may be debatable considering the existing

data set. The only well-dated Devonian paleomagnetic results from Devonian rocks for Avalon in the published literature are from a series of post-Acadian (Late Devonian) intrusives in eastern Newfoundland and these yield a mean inclination of 51°, $\alpha_{95}=10°$ (IRVING and STRONG, 1985). The Fourchu gabbro "A component" (Table 1) yields a site mean declination and inclination of 165°, 41°. Now, the declination and inclination (170°, 65°, $\alpha_{95}=10°$) of the Fourchu strata (FS) are not very different from the above mentioned Devonian directions for the Avalon terrane. Clearly the inclination (65°) of FS is steeper than a mean inclination of 51° but this is barely statistically significant. A steeper inclination could certainly indicate an older age of magnetization (e.g. Silurian or Ordovician times).

The characteristic direction of the FS (170°, +65°) and the B direction (312°, +63°) identified by JOHNSON and VAN DER VOO (1986) are both interpreted as secondary magnetizations, but they cannot have the same age as the declinations are different. One way to solve this discrepancy is to agree that the B component obtained by JOHNSON and VAN DER VOO (1986) coincides with the Late Hadrynian Avalonian orogeny (clearly this age is also subject to some uncertainty) and that the age of the FS component is younger. How much younger is the next question! As the characteristic direction of the FS component is different from recently reported paleomagnetic directions in Mid-Cambrian rocks from the Avalon terrane of Nova Scotia (JOHNSON and VAN DER VOO, 1986), it is thus unlikely to represent an overprint acquired during the Cambrian interval. Thus, the FS component is most likely of Ordovician or Silurian age (SEGUIN and RAO, 1989). A similar characteristic direction (155°, +61°) was obtained for Upper Ordovician-Lower Silurian rocks from Nova Scotia (VAN DER VOO and JOHNSON, 1985).

Finally, an independent comparison between the directions of JOHNSON and VAN DER VOO (1986) for the Fourchu strata and the magnetization direction observed in the gabbro (this study) may shed some light on the age of the B component of this study. The B component of magnetization is very similar to that of the hematite component identified by JOHNSON and VAN DER VOO (1986) in the Fourchu strata. This implies that these two components are of similar age and that the B component in the gabbro is probably a secondary magnetization older than the A component (Early Silurian). This statement is based on the value of the inclination (the older is steeper and the younger shallower).

After the different steps of thermal treatment, the magnetic susceptibility was measured to test the possibility of formation of an artificial magnetic memory carrier (Fig. 8). It is usually assumed that the formation of new magnetic minerals during heating will produce an observable change in magnetic susceptibility. An increase in susceptibility during thermal demagnetization is generally thought to indicate the formation of new magnetite, while a decrease in susceptibility often results from oxidation of (titano)-magnetite to (titano)hematite or inversion of (titano)maghemite to (titano)hematite (COLLINSON, 1983). The formation of new magnetite, which can result from the breakdown of clay minerals or other ferromagnesian silicates during heating (COLLINSON, 1983), is of greatest concern to paleomagnetic studies because such magnetite can acquire large spurious components of remanent magnetization during or after the heating which is commonly revealed by an increase in remanent intensity or erratic change of direction.

This test indicated that whenever the susceptibility remained relatively constant (FG.6-1.3, FG.6-2.1, FG.7-3.1, FG.6-5.1, Fig. 8) or decreased (FG.4-5.3, Fig. 8) in the

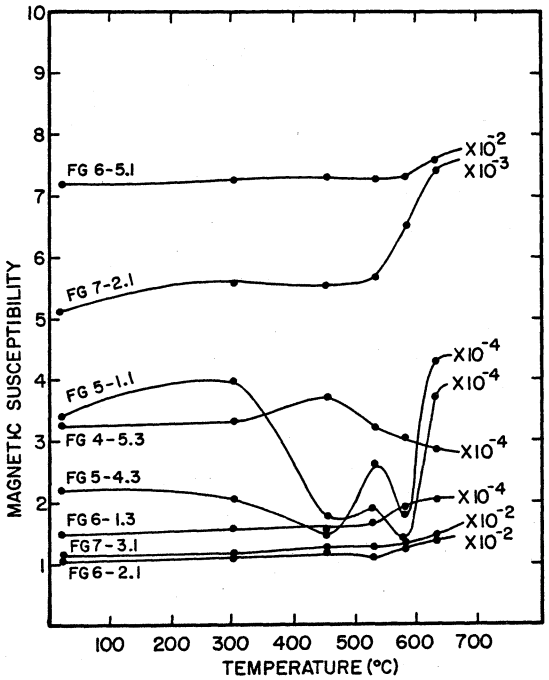


Fig. 8. Variation of volume magnetic susceptibilities (K) in typical and varying gabbro specimens as a function of temperature. Measurements of K after each thermal demagnetization step allows the determination of the rate and degree of oxidation in the 570–630° C range. This increase in susceptibility values cannot be caused by the presence of hematite or the oxidation of magnetite. It is explained by the presence of titanohematite. The K values are expressed in SI units.

samples throughout the entire treatment, only component A was isolated. When the susceptibility increases substantially at temperature higher than 570° C (FG.5-1.1, FG.5-4.3, FG.7-2.1, Fig. 8), component B is often detected. This test indicates that a fraction of component B is caused by the formation of a small amount of (titano)hematite. The magnetite formed may or may not be capable of carrying a stable remanence. It is noted that the direction of the B component does not change in the temperature range: 525°–630°. This indicates that the remanence measured above 570° C is a real component of magnetization carried by (titano)hematite and not a spurious laboratory component of magnetization. A similar phenomenon was observed in the Cretaceous gabbro from Mount Mégantic (SEGUIN and ST. HILAIRE, 1985). (Titano)hematite is occasionally found in gabbros; presence of (titano)hematite takes place at high temperature during initial cooling and is not necessarily due to weathering (MCELHINNY, 1973). In the reference cited, the titanohematite is intergrown with titanomagnetite suggesting that the oxidation is taking place at the same time as high temperature exsolution of Ti-rich and Ti-poor phases in primary (igneous) grains. As no such evidence was noted in the Fourchu gabbro, the occurrence of secondary (titano)hematite is more likely. JOHNSON and VAN DER VOO (1986) interpret this hematite component as being acquired during oxidation processes associated with uplift and subaerial exposure of the Fourchu Group following the latest Precambrian Avalonian orogen. For the above mentioned reasons,

component B is older than component A and it could be either secondary or primary. Component B (147, −54) is not significantly different from components interpreted as Late Precambrian (600–700 Ma) from Finn Hill ignimbrites (123, −40) (SEGUIN, 1985), porphyrites and rhyolites (131, −48) (SEGUIN and LORTIE, 1986), Calmer sandstones (108, −54) (IRVING and STRONG, 1985) in the Avalon terrane of Newfoundland and Harbour Main Group (317, −22) (SEGUIN and GIASSON, 1987).

Component A (165, +41) is presumably of Paleozoic age and its direction is comparable to Late Devonian directions obtained in the St. Lawrence granite (185, +50; IRVING and STRONG, 1985), and in the Harbour Main Group overprint (175, +35; SEGUIN and GIASSON, 1987) from the Avalon of Newfoundland and even the Ste-Cécile–St. Sébastien granite (152, +39; SEGUIN *et al.*, 1982b) from southern Quebec. In summary, component A is interpreted as a secondary component. Component B may be either primary or secondary and in this last event it spans the Late Precambrian–Late Devonian interval. Considering the above mentioned arguments in favour of a post-Late Hadrynian age for component B, we interpret this component as a secondary magnetization overprinted by component A.

8. Comparison with Other Paleomagnetic Data

The paleopoles of components A and B are: 45° W, 19° S (dp=3.4°, dm=5.8°) and 196° E, 63° N (dp=6.3°, dm=8.9°) respectively.

From similarity of directions, the age of magnetization assigned to component A is Late Devonian and Late Devonian to Late Precambrian for component B. For comparison with other Late Precambrian data, only the direction of component B is retained. The paleolatitude of component B is steeper than the paleolatitude of other Late Precambrian rocks from the Avalon Zone (SEGUIN, 1985; SEGUIN and LORTIE, 1986; SEGUIN and GIASSON, 1987; IRVING and STRONG, 1985) which range from 15° to 35° S. This suggests that the acquisition of the remanence of component B is of Early Paleozoic age or that Late Precambrian paleomagnetic directions were overprinted. Given the relatively good coincidence of paleolatitudinal data for the Late Precambrian period, an Early Paleozoic remagnetization for component B is more probable.

For this same Late Precambrian period, the paleolatitudes of cratonic North America are located close to the equator (5° to 20° S) (e.g. VAN DER VOO, 1982). Consequently, by Late Hadrynian time, the Avalon Terrane and North America were separated by a sea which was at the most 1500 to 2500 km wide. Considering the uncertainties associated with paleomagnetic poles and paleolatitudes, there is little ground for a 2000 km separation between cratonic North America and the Avalonian Terrane. If an ocean existed between cratonic North America and the Avalon Terrane at that time, it was certainly not wider than 500 to 1000 km. Alternatively, there is a possibility that Hadrynian magnetic signatures from both cratonic North America and Avalonian Terrane were remagnetized at a later data (Early Paleozoic). The separation between the two blocks remains undetected if the ocean separating the two blocks had closed by that time.

A comparison of the paleolatitudes of Avalon with those from North Africa (Gondwana) and northwestern Europe during this same period could eventually yield information about the provenance of the Avalon terrane. Few (6 poles) and poorly constrained paleomagnetic data are available from northwestern Europe (BRIDEN and

DUFF, 1981; PERIGO *et al.*, 1983) in the −600 to −650 Ma interval. The paleolatitudes obtained from these paleomagnetic data, assuming BULLARD *et al.* (1965) fit between Avalon and northwestern Europe are clustered around the equator. Using a similar procedure for the limited number (4–5) of Late Hadrynian Gondwana poles (data listed in HAGSTRUM *et al.*, 1980; PERIGO *et al.*, 1983), the paleolatitudes obtained range from 20 to 40° S.

9. Tectonic Significance of the Results

The existing paleomagnetic results suggest that in Late Hadrynian time, the Avalon microcontinent and cratonic North America did not occupy significantly different latitudes and were not widely separated. In Early Paleozoic times, the paleolatitude obtained from component B indicates that the Avalonian Terrane was situated some 15° S of cratonic North America. Other possible candidates for the provenance of Avalon in the Atlantic borderlands include North Africa and northwestern Europe. North Africa belongs to the Gondwana cratonic plate and northwestern Europe to the hypothetical Armorica plate (VAN DER VOO, 1982). A comparison of the paleolatitudes and paleopoles of Avalon and northwestern Europe suggests that Avalon is unrelated to northwestern Europe during the Hadrynian period. A similar comparison of Avalon with Gondwana clearly shows that a reconstruction of the two continents for the same time interval is highly probable on paleomagnetic ground. The proposed paleoreconstitution of these continents favours the tectonic models proposed by STRONG (1979), PIQUÉ (1981) and O'BRIEN *et al.* (1983). Obviously, this reconstruction is not exclusive; other alternatives are still acceptable, e.g. possibility of an independent miniplate. Late Precambrian paleomagnetic data for Avalon are sparse and consequently the geophysical tests for the provenance of Avalon awaits additional reliable paleomagnetic data.

The timing for the docking of the Avalon terrane unto cratonic North America is much better documented. Significantly similar directions of magnetization for the Late Devonian period obtained in the Avalon terrane, the Andean margin of North America and cratonic North America clearly indicate that Avalon was already accreted to cratonic North America by the end of the Acadian orogeny (−365 Ma). The relative motion of the Avalon terrane in the −600 to −365 Ma interval is still ill-defined except possibly for the Devonian period (SEGUIN, 1989). An Early to Middle Devonian time appears to be the most probable period for the accretion of Avalon to cratonic North America. In view of the small size of the gabbros, and the evidence that the area investigated was affected by at least two orogenic episodes which folded the Fourchu strata after they were intruded, local tectonic rotation of the gabbros seems to be rather unprobable; indeed, there are few field indications for such rotation. As one of the assigned ages (component B) is older than the orogenic episodes, there is a small but real possibility that this direction of magnetization in the gabbros may not be representative of the dipole field direction for the Avalonian Terrane at any period. Still, the similarity of the poles from this study with other existing paleopoles suggest that the amount of rotational motion in the gabbro is small. In addition, there is little difference in the directions of magnetization between sites FG-1 to 4 and FG-6 and 7 which are over 10 km apart although they are situated subparallel to the folding axis; this minimizes the possibility of a local rotation.

10. Conclusions

The present paleomagnetic study is centered on three Fourchu gabbro intrusive bodies from the Avalon Zone of Nova Scotia. The same components of magnetization were obtained for the three bodies. The fact that the separate gabbro bodies sampled yield the same direction (and in particular for the B component) may be cited as evidence for a lack of local tectonic rotations; however, the same data could correctly be interpreted as evidence that the magnetization in all three bodies is, in fact, secondary. Two definite components (A and B) were isolated. Component B (147, −54) is interpreted as a probable secondary magnetization corresponding to the uplift and subaerial exposure of the gabbros (Cambro-Ordovician age). Component A (165, +41) is interpreted as a Siluro-Devonian overprint. It is then concluded that the age of magnetization of the three intrusions is approximately synchronous. A comparison of paleolatitudes from Late Hadrynian paleomagnetic data from Avalon terrane(s), cratonic North America, Gondwana (North Africa) and northwestern Europe suggest that: 1) North America was not far away from Avalon Zone, 2) the Avalon Zone was probably part of North Africa and 3) at Late Devonian time or slightly before (Middle Devonian) the Acadian orogeny, the Avalon terrane docked unto North America. As no Ordo-Silurian Taconic orogeny is recognized in the Avalon Zone, the Devonian period appears to be the earliest time at which this collision occurred.

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